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Network Layer

The **network layer** (**Layer 3**) of **OSI** controls the exchange of data packets, as these cannot be directly routed to the receiver and therefore have to be provided with routing nodes. The data packets are then transferred from node to node until they reach their target. To implement this, the **network layer** identifies the individual network nodes, sets up and clears connection channels, and takes care of routing and data flow control. When sending the packets, addresses are evaluated, and the data is routed through the network from node to node. There is usually no processing of the data in the layers above the **L3** in the nodes. Based on the addresses, the routing and the construction of routing tables are done.

In short, it is responsible for the following functions:

- **Logical Addressing**
- **Routing**

Protocols are defined in each layer of **OSI** , and these protocols represent a collection of rules for communication in the respective layer. They are transparent to the protocols of the layers above or

below. Some protocols fulfill tasks of several layers and extend over two or more layers. The most used protocols on this layer are:

- IPv4 / IPv6
- IPsec
- ICMP
- IGMP
- RIP
- OSPF

It ensures the routing of packets from source to destination within or outside a subnet. These two subnets may have different addressing schemes or incompatible addressing types. In both cases, the data transmission in each case goes through the entire communication network and includes routing between the network nodes. Since direct communication between the sender and the receiver is not always possible due to the different subnets, packets must be forwarded from nodes (routers) that are on the way. Forwarded packets do not reach the higher layers but are assigned a new intermediate destination and sent to the next node.



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